

2 Weighted Voting

2.2 Banzhaf Power Index

Definition 2.2.1 To calculate the Banzhaf Power Index:

- List all winning coalitions.
- In each winning coalition, determine all players who are critical.
- Count up how many times each player is critical.
- Convert these to fractions by dividing by the total number of times any player is critical.

Example 2.2.2 Calculate the Banzhaf Power Index for the three players in the following weighted voting system

$$[10 : 7, 6, 2].$$

Example 2.2.3 Calculate the Banzhaf Power Index for the following weighted voting system

$$[8 : 6, 3, 2].$$

$$P_1 = 3/5, P_2 = 1/5 \text{ and } P_3 = 1/5.$$

Example 2.2.4 An executive board consists of a president (P) and three vice-presidents (V1,V2,V3). For a motion to pass it must have three yes votes, one of which must be the president's. Find a weighted voting system to represent this situation, then find the Banzhaf power index of the system. The list of all coalitions is given below

$$\begin{array}{cccc} \{P_1\} & \{P_2\} & \{P_3\} & \{P_4\} \\ \{P_1, P_2\} & \{P_1, P_3\} & \{P_1, P_4\} & \{P_2, P_3\} \\ \{P_2, P_4\} & \{P_3, P_4\} & \{P_1, P_2, P_3\} & \{P_1, P_3, P_4\} \\ \{P_1, P_2, P_4\} & \{P_2, P_3, P_4\} & \{P_1, P_2, P_3, P_4\} & \end{array}$$

One example of this is $[8 : 4, 2, 2, 2]$. The Banzhaf power index of this weighted voting system is

$$P_1 : 2/5 \ P_2 : 1/5 \ P_3 : 1/5 \ P_4 : 1/5$$

Example 2.2.5 Compute the Banzhaf power index for the following weighted voting system.

$$[16 : 7, 6, 3, 3, 2].$$

The winning coalitions are listed below:

$$\begin{aligned} &\{P_1, P_2, P_3\}, \{P_1, P_2, P_4\}, \{P_1, P_2, P_3, P_4\}, \\ &\qquad \{P_1, P_2, P_3, P_5\}, \{P_1, P_2, P_4, P_5\}, \\ &\qquad \qquad \{P_1, P_2, P_3, P_4, P_5\} \end{aligned}$$

Power index for this example is
$P_1 : 3/8 \quad P_2 : 3/8 \quad P_3 : 1/8$
$P_4 : 1/8 \quad P_5 : 0$